

Architecture Selection — Pharmacy Expiry & Re-order Dispatch Engine

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Problem Statement: #15 **Domain:** Healthcare & Telemedicine **Course:** Software Engineering Lab 3

Architecture Selection: "We chose Layered Architecture for the Pharmacy Expiry & Re-order Dispatch Engine."

1. Architectural Choice

We selected Layered Architecture for the Pharmacy Expiry & Re-order Dispatch Engine. The system is partitioned into separate horizontal layers: the Presentation Layer (Pharmacy User Interface), the Business/Service Layer (Order Manager Component, Inventory & FEFO Manager Component, Re-order / Purchase Order Service, and Payment Service), and the Data Layer (Inventory Database Component), with clear boundaries to external actors such as the Inventory Supplier. Each layer has specific responsibilities and communicates strictly with adjacent tiers through clearly defined interfaces.

2. Architectural Justification: Key Reasons

Reason 1: Clear Separation of Concerns. The Layered Architecture establishes a clean division of responsibilities across the user interface, business logic, and database. The Pharmacy User Interface is solely responsible for rendering stock screens, displaying near-expiry warnings, and capturing clerk actions. The service layer houses all inventory business rules, while the database component is dedicated to persistent storage and transactional integrity. This separation ensures that interface modifications do not affect core pharmacy algorithms and schema updates remain isolated from clerk workflows.

Reason 2: Independent Maintainability of FEFO Logic and Stock Re-ordering. In a hospital pharmacy, rules governing First-Expired-First-Out (FEFO) dispensing sequences, 30-day near-expiry monitoring, and safety-stock threshold triggers evolve independently to meet institutional guidelines. Because the Inventory & FEFO Manager Component and the Re-order / Purchase Order Service reside in dedicated business service components, their picking logic and threshold algorithms can be maintained, tested, or modified independently without tightly coupling the presentation layer and data layer.

3. Security Advantage

Layered separation provides controlled access to sensitive medicine inventory, batch histories, and supplier procurement information. Rather than permitting the Pharmacy User Interface to access or query the database directly, all operations are mediated through business service components. The service tier validates clerk identity, checks authorizations, enforces dispensing constraints, and records transactions in the immutable audit log before executing queries against the Inventory Database, preventing direct client-side database exposure and unauthorized adjustments.

4. Performance Benefit

Frequently used business operations—such as real-time inventory lookup, FEFO batch selection, and stock-threshold checking—are handled directly within dedicated service components. By filtering and sorting batches in the application services, the system avoids redundant database round-trips and executes efficient, consolidated queries. Furthermore, decoupling automated purchase order dispatch into an independent service ensures supplier communication does not block interactive dispensing checkout transactions at the clerk's terminal.